

SHAURYA PUROHIT

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PROFESSIONAL SUMMARY

I am proficient and have extensive hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, Federated Learning, and Data Analytics. My expertise spans data preprocessing, feature engineering, model prototyping, evaluation, and deployment at scale using modern ML frameworks. I have good knowledge of core machine learning techniques such as classification, regression, and clustering, with a strong foundation in Python, TensorFlow, Keras, and related data science tools, and experience in building and optimizing ML pipelines for large datasets. In addition to my work in ML, I have applied these skills to domains such as cybersecurity and anomaly detection for critical infrastructure.

EDUCATION

Pursuing **PhD in Computer Engineering**
Iowa State University - Ames, Iowa
CGPA: 3.86/4.00

August 2019 – February 2025

Master of Engineering in Computer Engineering (Minor in Statistics)
Iowa State University - Ames, Iowa
CGPA: 3.84/4.00

August 2019 - May 2022

Bachelor of Technology in Information Technology
Manipal University Jaipur - Jaipur, India
CGPA: 9.58/10.00

August 2015 - May 2019

PROFESSIONAL EXPERIENCE

Eaton – Golden, CO, USA

March 2025 – Present

Lead Engineer

- Working on advanced computational research on government-funded projects, including the development of anomaly detection machine learning algorithms for storm prediction (with >90% accuracy), pre-storm preparation control strategies, and optimization of dispatch and network reconfiguration in islanded mode to enhance grid resilience by up to 30%.

Iowa State University - Ames, Iowa

Graduate Research Assistant

May 2020 – February 2025

- Worked on two projects in collaboration with Argonne National Laboratory
 - The first project enhances real-time operations of distributed energy resources, providing cost-effective, real-time cybersecurity for individual DERs within an aggregation pool through Reinforcement learning and Game theory.
 - The second bolsters grid cybersecurity through a novel Machine and Federated Learning methodology (with accuracy >96%). The project underscores a commitment to integrating Machine Learning to address complex energy and security challenges through threat intelligence and anomaly detection.
- Worked on Cybersecurity of DER communication protocols and EV Charging using Federated Learning.
 - Developed federated learning anomaly detection models enhancing data privacy and cybersecurity in DER and EV charging networks, achieving high accuracy and reduced loss on publicly available large datasets.
- Had the honor of being **selected as a senior PhD student to serve on a panel at the CAE-R Symposium 2023**, where I presented both my research and poster.
- Worked on Game-theoretic Optimization of Cybersecurity Investment Strategies for EV Charging Stations using Attack Defense Trees.
 - This framework integrates the CIA triad and MITRE ATT&CK framework to analyze attack paths, optimize resource allocation, and formulate robust defense strategies using TTPs.
- Worked on cybersecurity of autonomous vehicles using Machine Learning & Deep Neural Networks
 - Developed a Hybrid Anomaly detection system for preventing cyber-attacks on intra-vehicular communication networks with the help of rule based and machine learning techniques. Achieved an accuracy of around 95%.
 - Submitted the enhanced journal version of this work using Deep Neural Networks and variety of datasets. Achieved an accuracy of more than 97% and better latency.
- Worked on a project titled Cloud-based multi-sensor remote data acquisition system.

Mutual of Omaha – Nebraska, USA

May 2022 – December 2022

Information Technology Intern

- Worked as an I/S intern with Security and ML team.
 - Worked on Deep Neural Networks for Forcepoint Framework.
 - Worked on complete automation and full stack management of an in-house project titled “Moonet” using Jenkins CI/CD pipeline architecture.

Iowa State University - Ames, Iowa

Graduate Teaching Assistant

August 2021 - December 2021

- Worked as a TA for a 500-level course: Real-Time systems (**CPR E 558**)
- Responsibilities included: one-to-one interaction with students for solving queries, helping students in their final course project, providing feedback by grading assignments, quizzes, and exams.

Graduate Teaching Assistant

August 2019 - May 2020

- Worked as a TA for courses: Introduction to Computer Engineering and Problem Solving (**CPR E 185**) & Problem Solving in Software Engineering (**S E 185**).
- Responsible for conducting labs every week to help students write programs in C, helping students in their final course project, holding office hours to facilitate one-to-one interaction for resolving queries, grading homework, lab reports, exams and providing appropriate feedback.

ValueLabs – Hyderabad, India

January 2019 - April 2019

Software Engineer Intern

- Worked as a Full-Stack Developer on a real-time client project titled MfExpert (Micro- Finance Expert) and designed a web application using Angular 7, Node.js and PostgreSQL.

IIT BHU – Varanasi, India

May 2018 - July 2018

Machine Learning Intern

- Worked on a project titled Autonomous Car Detection using Deep Learning under the guidance of Dr. Tanima Dutta.
- Implemented Car Detection in images and videos using Machine Learning and Deep Learning and achieved a high accuracy of 98%.

CERTIFICATIONS

- Machine Learning
Issued by: Stanford University (via Coursera)
- Cybersecurity Specialization
Issued by: IBM (via Coursera)
- Cybersecurity Fundamentals
Issued by: IBM
- AWS Machine Learning
Issued by: Amazon Web Services (via Coursera)
- Programming in Java
Issued by: NIIT Technologies

PUBLICATIONS

Conferences:

- **S. Purohit** and M. Govindarasu, "ML-based Anomaly Detection for Intra-Vehicular CAN-bus Networks," 2022 IEEE International Conference on Cyber Security and Resilience (CSR), Rhodes, Greece, 2022, pp. 233-238, doi: 10.1109/CSR54599.2022.9850292. (**Published**)
- **S. Purohit** and M. Govindarasu, "Cybersecurity Investment Analysis for Electric Vehicle Charging Infrastructures," 2023 Resilience Week (RWS), National Harbor, MD, USA, 2023, pp. 1-6, doi: 10.1109/RWS58133.2023.10284670. (**Published**)
- **S. Purohit** and M. Govindarasu, "FL-EVCS: Federated Learning based Anomaly Detection for EV Charging Ecosystem," 2024 33rd International Conference on Computer Communications and Networks (ICCCN), Kailua-Kona, HI, USA, 2024, pp. 1-9, doi: 10.1109/ICCCN61486.2024.10637543. (**Published**)
- **S. Purohit** and M. Govindarasu, "Federated Learning based Anomaly Detection for Distributed Energy Resource Communication," 2024 IEEE Power & Energy Society General Meeting (PESGM), Seattle, WA, USA, 2024, pp. 1-5, doi: 10.1109/PESGM51994.2024.10688634. (**Published**)
- B. Blakely, **S. Purohit**, K. Ogbuefi, I. Prieto and M. Govindarasu, "A Federated Intrusion Response Network for Grids," IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society, Chicago, IL, USA, 2024, pp. 1-4, doi: 10.1109/IECON55916.2024.10905821. (**Published**)
- P. K. Mallaiah, N. Saqib, **S. Purohit** and M. Govindarasu, "ML-based Multiclass Anomaly Detection System for Charge Profile Manipulation Attacks in EV Charging Ecosystem," 2025 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), North York, ON, Canada, 2025, pp. 1-6, doi: 10.1109/SmartGridComm65349.2025.11204647. (**Published**)
- K. Wang, **S. Purohit** and M. Govindarasu, "A Robust Framework for Cyber Attack Detection and Mitigation in DER Networks using Reinforcement Learning and Game Theory, " 2025 IEEE Power & Energy Society General Meeting (PESGM) (**Accepted**)

Journals:

- **S. Purohit**, M. Govindarasu and B. Blakely, "FL-ADS: Federated Learning Anomaly Detection System for Distributed Energy Resource Networks", " IET Cyber-Phys. Syst., Theory Appl. e70001(2025), doi: 10.1049/cps2.7000118. (**Published**)
- **S. Purohit** and M. Govindarasu, "HAVEN: A Hybrid Anomaly Detection System for Intra-Vehicular CAN-bus Communication using Rule-based and Neural Networks" , IEEE Transactions on Dependable and Secure Computing. (**Submitted- Under Review**)

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, R
- **Web Based Technologies:** HTML, CSS, Angular, Node.js, PHP, JavaScript, jQuery, Ajax, XML, DHTML
- **ML Frameworks:** TensorFlow, Keras, Pandas, NumPy, Scikit, PyTorch, Flower
- **Cloud Platforms:** AWS, Elastic Cloud
- **Subject specific software:** Docker, RStudio, STAR UML, AutoCAD
- **Databases:** MySQL, PostgreSQL

ACHIEVEMENTS & AWARDS

- **Graduate Research Excellence Award**, Iowa State University, 2025 (for outstanding research contributions during PhD studies).
- Secured **1st Rank in B. Tech (IT) 3rd year & 4th year** for the academic year 2017-18 and 2018-19 respectively.
- Secured **Statewide Ranking 1st & India Wide Ranking 3rd** in the Wheebox Employability Skills Test in 2017.
- Published article for e-magazine **EasyShiksha** in 2018.